Unified Biquaternion Theory (UBT)

Textbook and guide for engineers

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Contents

Preface: how to read this textbook

This book is a Unified workbook and reference text Biquaternion Theory (UBT). Her first task is to pass the math so much comprehensible and coherent so that it is possible to reconstruct again later the links between by individual UBT branches without relying on memory or old chats. That's why they are in classical mathematical results are also deliberately derived from the text, if they are needed for the big picture of UBT, even though the math itself may be decades or hundreds of years old known

The textbook distinguishes three layers:

1. **classical foundations**: standard maths and physics with sufficient by inference to make the text as self-sufficient as possible;
2. **UBT bridges**: controlled conversions of these standard structures into UBT notation, constrained models or specific hypotheses;
3. **UBT statements and open gaps**: canonical field dependent statements UBT, its action, dynamics or empirical program.

Classic identity does not become a new result just because UBT uses it. Likewise, the fact that it is well-known mathematics does not mean that it is from a textbook throw out: the research value may lie in how known structures are folded together and what new question for UBT their union precisely formulates. Canonical claim papers therefore remain conservative, while the textbook can be broad and didactic if the statuses of the claims are always distinct.

A live textbook must not take active theory from `ARCHIVE/`. Current status is first read from authoritative registers, especially `STATUS.md`, `WHAT_IS_PROVED.md` and `CLAIMS.yaml`; only then should it use active math from `canonical/`. Standard mathematics can be found in the textbook derive directly. Historical files only serve to describe the history of an idea. The migration map is in `docs/textbook/SOURCE_MAP.md`.

The primary student edition is in Czech. The English version may later be run as parallel editions with the same mathematical structure and the same assertion statuses; new chapters should not be written as a random mixture of Czech and English.

Each live source file also uses machine-readable AI-provenance repository policy. This textbook is Tier C work material until human the editor does not explicitly change the tier in the authoritative provenance map; The AI agent must not carry out own promotion or certification.

The derivations in the papers and the theoretically important parts of the textbook are given independently check against `docs/DERIVATION_VERIFICATION_POLICY.md`. For formalizable exact mathematics is preferred Lean; are used next to it independent CAS and numerical controls. However, it only verifies a successful calculation in the tool what was actually coded into the tool and does not in itself change the physical the status of the UBT claim.

Part I

Foundations and geometry

Chapter 1

UBT per hour: motivation, architecture and current status

1.1 What is this chapter for?

Unified Biquaternion Theory (UBT) asks whether spacetime geometry and internal field structure can be encoded in one complex-quaternion-valued master field, rather than introduced as unrelated fundamental objects. The active starting notation is

$$\Theta(q, \tau) \in \mathbb{B} = \mathbb{C} \otimes \mathbb{H}, \qquad \tau = t + i\psi.$$

The notation is compact, but it hides several logically different questions: what is defined, what is kinematically representable, what follows from an action, and what is only a research hypothesis. The textbook keeps those levels separate throughout.

1.2 The current local geometry string

The clean local GR branch starts from the covariant derivative of the single field,

$$E_\mu := \mathcal{N}_0^{-1/2} D_\mu \Theta.$$

On the real Lorentz slice, write

$$E_\mu = i e_\mu{}^0 \mathbf{1} + e_\mu{}^k \mathbf{e}_k, \qquad e_\mu{}^a \in \mathbb{R}.$$

Quaternion conjugation is denoted by $\sharp$. The symmetrised product is central and defines the metric,

$$\boxed{\frac{1}{2}\left(E_\mu^\sharp E_\nu + E_\nu^\sharp E_\mu\right) = g_{\mu\nu}\mathbf{1}.}$$

Thus the minimal local metric construction does not require an observer projection, a compact-fibre average, or an embedding map. This replaces an older architectural picture preserved in the historical archive.

The same first derivative behaves as a tetrad. For a nondegenerate tetrad the map $e_\mu{}^a \mapsto g_{\mu\nu}$ has the familiar counting

$$16 - 6 = 10,$$

because six local Lorentz changes leave the ten metric components invariant. This resolves the local rank problem at the kinematic level.

1.3 Connections, curvature and what is still dynamic

A local Lorentz/biquaternionic connection enters D_μ. In a torsion-free GR branch the compatible spin connection is the Levi–Civita spin connection computed from the tetrad. With torsion,

$$\omega = \mathring{\omega}(e) + K(T),$$

so specifying the tetrad and torsion determines the metric-compatible connection. Curvature belongs to the commutator of covariant derivatives, not to the algebraic commutator of the tetrad legs:

$$\mathcal{R}_{\mu\nu} := [D_\mu, D_\nu].$$

The present GR work closes important local representability questions, but it does not yet derive all Einstein dynamics from the original UBT master action. In particular, action selection, physical torsion constraints, the full perturbation bridge, and global continuation remain status-controlled open problems. The authoritative wording is maintained in `STATUS.md` and `WHAT_IS_PROVED.md`.

1.4 Complex time and theta/spectral branch

The canonical notation retains

$$\tau = t + i\psi,$$

with ψ interpreted in the active complex-time branch as an internal phase coordinate rather than a second macroscopic clock. Reduced theta models then naturally contain factors of the form

$$S(t, \psi) = \sum_n a_n e^{-\pi\psi n^2} e^{i\pi t n^2}.$$

This one expression admits several standard representations: Fourier in t, Laplace or Mellin in ψ, Jacobi/Poisson duality, and, in controlled quadratic examples, a relation between mode sums and path/winding sums. Those classical tools are developed explicitly later because they organise several otherwise disconnected UBT research tracks.

1.5 Calibration and particle sectors

The algebra $\mathbb{C} \otimes \mathbb{H} \cong M_2(\mathbb{C})$ naturally supports left and right actions, spinorial Lorentz transformations, and internal symmetry constructions. Some gauge-sector statements are proved or conditional at stated levels; others, including parts of hypercharge, generation selection, and dynamical symmetry breaking, remain open. This textbook therefore never infers a full Standard Model derivation merely from the existence of the biquaternion algebra.

1.6 The fine structure constant: a current safe assertion

The alpha programme contains a reproducible structural mechanism selecting the integer 137 under stated assumptions, including a prime-stability result. However the overall track is currently not a first-principles derivation of the physical fine-structure constant. Two blockers are especially important:

1. the effective coefficient required by the winding potential has not been derived uniquely from the UBT action (Gap G137-B);

2. the audited loop multiplicity and the twist-sector multiplicity are not yet identified by a proved bridge.

Consequently neither the phrase “fit-free alpha derivation” nor the numerical value 137.036... should be presented here as a closed UBT prediction. Chapter 7 derives the conditional mathematics and states the gaps at every step.

1.7 How to read status tags

The repository uses two independent bookkeeping systems that must not be confused.

First, scientific claim levels distinguish standard identities, proved UBT lemmas, conditional results, motivated conjectures, observations, and open gaps. Second, AI provenance tiers record editorial/review status of files. The latter are defined by `PROVENANCE_TIERS.yaml`: Tier A is author-attested, Tier B machine-verified, Tier C working, and Tier D historical. This textbook is Tier C working material. Its role is to explain and connect the mathematics, not to silently promote a claim to Tier A.

1.8 Authoritative source rule for a textbook

The source map in `docs/textbook/SOURCE_MAP.md` records where each chapter gets its live status. Historical material in `ARCHIVE/` can be valuable for understanding how an idea evolved, but it is never silently `\input` as the current theory.

Chapter 2

Mathematical basis: biquaternions without tears

This chapter introduces the algebra needed by later chapters. Detailed Lorentz geometry is derived self-contained in Chapter 3; the live textbook does not import the historical consolidation archive.

Chapter 3

Lorentz geometry on the Hermitian section $\mathbb{H}_{\mathbb{C}}$

3.1 Classic basis: $\mathrm{Herm}(2, \mathbb{C})$

This section is standard spinor/Lorentz mathematics. It is included because UBT uses the same algebra through the isomorphism

$$\mathbb{B} = \mathbb{C} \otimes \mathbb{H} \cong M_2(\mathbb{C}).$$

Let σ_i be the Pauli matrices and associate a real four-vector $x = (x^0, x^1, x^2, x^3)$ with the Hermitian matrix

$$X = x^0 I_2 + x^i \sigma_i = \begin{pmatrix} x^0 + x^3 & x^1 - ix^2 \\ x^1 + ix^2 & x^0 - x^3 \end{pmatrix}.$$

A direct determinant calculation gives

$$\boxed{\det X = (x^0)^2 - (x^1)^2 - (x^2)^2 - (x^3)^2.}$$

Thus the determinant on the Hermitian slice is exactly the Minkowski quadratic form with signature $(+ - --)$.

3.2 Action $SL(2, \mathbb{C})$

For $A \in SL(2, \mathbb{C})$ define

$$X \longmapsto X' = AXA^\dagger.$$

Hermiticity is preserved because

$$(AXA^\dagger)^\dagger = AXA^\dagger.$$

The determinant is also preserved:

$$\det X' = \det A \, \det X \, \det A^\dagger = \det X,$$

since $\det A = 1$ and $\det A^\dagger = \overline{\det A} = 1$. Therefore the induced real linear transformation of the four coefficients preserves the Minkowski norm. Restricting continuously to the identity gives a homomorphism

$$SL(2, \mathbb{C}) \longrightarrow SO^+(1, 3).$$

Both A and $-A$ act identically on X, so the kernel contains $\{\pm I_2\}$. The standard result is the double cover

$$\boxed{SL(2, \mathbb{C})/\{\pm I_2\} \cong SO^+(1, 3).}$$

No split-quaternion algebra is required for this construction.

3.3 Conversion to biquaternion notation

Use the canonical matrix representation

$$\mathbf{e}_i \longmapsto i\sigma_i.$$

Then $\sigma_i \leftrightarrow -i\mathbf{e}_i$, and the Hermitian matrix above corresponds to the biquaternion

$$X_{\mathbb{B}} = x^0\mathbf{1} - ix^i\mathbf{e}_i.$$

This is simply another coordinate description of the same $\mathrm{Herm}(2,\mathbb{C})$ slice.

UBT's current tetrad convention often uses instead the Lorentz-real basis

$$\mathbf{u}_0 = i\mathbf{1}, \qquad \mathbf{u}_k = \mathbf{e}_k,$$

so a tetrad leg is

$$E_\mu = ie_\mu{}^0\mathbf{1} + e_\mu{}^k\mathbf{e}_k.$$

With quaternion conjugation $\sharp$,

$$E_\mu^\sharp = ie_\mu{}^0\mathbf{1} - e_\mu{}^k\mathbf{e}_k.$$

Using $\mathbf{e}_j\mathbf{e}_k + \mathbf{e}_k\mathbf{e}_j = -2\delta_{jk}$, one obtains

$$\begin{aligned}\frac{1}{2}(E_\mu^\sharp E_\nu + E_\nu^\sharp E_\mu) &= \left(-e_\mu{}^0e_\nu{}^0 + e_\mu{}^ke_\nu{}^k\right)\mathbf{1}\\ &= g_{\mu\nu}\mathbf{1}.\end{aligned}$$

This is the same Lorentz geometry with the opposite overall signature convention $(-+++)$. A signature convention is bookkeeping; it is not a new physical degree of freedom.

3.4 Why it matters to UBT

The classical result establishes three useful facts before any UBT dynamics is introduced:

1. complexified quaternions already carry the spinorial Lorentz action;
2. a four-dimensional Lorentzian quadratic form can be read internally from the algebra;
3. left/right multiplication matters, because the Lorentz action is naturally two-sided in matrix language.

What this classical construction does*not* prove is equally important. It does not prove that the UBT master field dynamically generates every desired tetrad, it does not derive Einstein's equations, and it does not select the physical connection. Those are UBT-specific bridge problems addressed in the covariant-tetrad chapter and the canonical GR closure ledger.

3.5 Specific boost as a control example

Take a boost of rapidity χ along the third spatial axis,

$$A = \exp\left(\frac{\chi}{2}\sigma_3\right) = \begin{pmatrix} e^{\chi/2} & 0 \\ 0 & e^{-\chi/2} \end{pmatrix}.$$

Applying $X' = AXA^\dagger$ and comparing coefficients gives

$$\begin{pmatrix} x'^0 \\ x'^3 \end{pmatrix} = \begin{pmatrix} \cosh\chi & \sinh\chi \\ \sinh\chi & \cosh\chi \end{pmatrix}\begin{pmatrix} x^0 \\ x^3 \end{pmatrix}, \qquad x'^1 = x^1, \quad x'^2 = x^2.$$

Then

$$(x'^0)^2 - (x'^3)^2 = (x^0)^2 - (x^3)^2,$$

which is the ordinary Lorentz boost written as an internal algebra action.

3.6 Living sources and historical note

The algebra convention is kept in `canonical/algebra/biquaternion_algebra.tex`; current GR use of the Lorentz-real tetrad is documented in the active GR track. The former `appendix_P6_lorentz_in_HC.tex` survives in the historical archive, but the live textbook no longer imports it.

Chapter 4

Covariant derivation, connections and pure UBT geometry

This chapter explains the geometric core of UBT in the form closest to the original intuition of the theory:

$$\Theta \quad \longrightarrow \quad E_\mu := D_\mu\Theta \quad \longrightarrow \quad \{E_\mu, E_\nu\}_\sharp,\ [D_\mu, D_\nu].$$

The anticommutator carries the metric, while the commutator of covariant derivatives carries curvature. No embedding map, compact-fiber average, or observer projection is needed in the minimal local construction.

4.1 Ordinary, partial and covariant derivatives

For an ordinary scalar function $f(x)$, the derivative df/dx measures how the number f changes when x changes. For a field on spacetime we use partial derivatives,

$$\partial_\mu\Theta := \frac{\partial\Theta}{\partial x^\mu}, \qquad \mu = 0, 1, 2, 3.$$

A partial derivative compares components written in a fixed basis. This is sufficient if the same basis can be used everywhere. Geometry becomes subtler when the local frame itself rotates or boosts from point to point.

A simple analogy is a vector written in a rotating coordinate frame. Even a physically constant vector acquires changing components because the basis is changing. The covariant derivative corrects for this basis change:

$$D_\mu\Theta = \partial_\mu\Theta + \rho_*(\Omega_\mu)\Theta.$$

Here Ω_μ is the connection and ρ_* tells us how the connection acts in the representation of Θ. For biquaternions, a left action, a right action, and a two-sided action are genuinely different and must not be silently interchanged.

4.2 What is a connection?

A connection is not a force by itself. It is a rule for comparing local frames at neighboring points. The symbol Ω_μ is the UBT internal or biquaternionic-frame connection.

Christopher symbols,

$$\Gamma^\rho{}_{\mu\nu},$$

are closely related, but they act on coordinate indices. For example,

$$\nabla_\mu V^\rho = \partial_\mu V^\rho + \Gamma^\rho{}_{\mu\sigma}V^\sigma.$$

Thus:

- $\Gamma^\rho{}_{\mu\nu}$ tells us how the coordinate basis changes;
- $\omega_\mu{}^a{}_b$, and its spin/biquaternionic lift Ω_μ, tell us how the local Lorentz frame changes.

They are related by the tetrad compatibility identity

$$\partial_\mu e_\nu{}^a - \Gamma^\rho{}_{\mu\nu} e_\rho{}^a + \omega_\mu{}^a{}_b e_\nu{}^b = 0.$$

They are therefore two descriptions of the same geometric transport, not the same coefficients. In particular,$\Gamma = \mathrm{Re}\,\Omega$ is not a valid identification.

4.3 Why the classic connection is not another free field

For a nondegenerate tetrad, metric compatibility and zero torsion determine a unique Lorentz connection:

$$\boxed{\mathring{\omega}_\mu{}^a{}_b = e_b{}^\nu \left(\mathring{\Gamma}^\rho{}_{\mu\nu}(g) e_\rho{}^a - \partial_\mu e_\nu{}^a \right).}$$

The circle marks the torsion-free Levi–Civita connection. Thus, in the ordinary torsion-free GR branch, the frame connection is computed from the tetrad just as the Christoffel symbols are computed from the metric. A local Lorentz rotation changes the numerical coefficients of the tetrad and the connection but does not change the geometry.

4.3.1 Track and counter

The torsion two-form is

$$T^a = de^a + \omega^a{}_b \wedge e^b.$$

A general metric-compatible connection can be written

$$\boxed{\omega = \mathring{\omega}(e) + K(T),}$$

where K is the contorsion. With $T^a = \frac{1}{2} T^a{}_{bc} e^b \wedge e^c$ and the convention above,

$$\boxed{K_{abc} = \frac{1}{2} \left(T_{cab} - T_{abc} - T_{bca} \right).}$$

This formula has an important conceptual consequence: once the tetrad and the torsion are specified, the metric-compatible connection is unique. The open full-UBT question is therefore not an arbitrary leftover Ω_μ; it is the dynamical law selecting T.

If the UBT vacuum equations imply $T = 0$, the connection reduces to the Levi–Civita spin connection. If spin or another biquaternionic source creates torsion, the same theorem reconstructs the connection from that torsion.

4.4 Flat space and explicit UBT representative

In flat Minkowski spacetime, using inertial Cartesian coordinates and a constant local frame, one can choose

$$\Gamma^\rho{}_{\mu\nu} = 0, \qquad \Omega_\mu = 0, \qquad D_\mu = \partial_\mu.$$

This statement can be strengthened from a choice of gauge to an explicit single-field represent. Easy

$$\boxed{\Theta_{\mathrm{SR}}(x) = \Theta_0 + \sqrt{\mathcal{N}_0} \left(i x^0 \mathbf{1} + x^k \mathbf{e}_k \right).}$$

Then

$$\mathcal{N}_0^{-1/2} \partial_0 \Theta_{\mathrm{SR}} = i\mathbf{1}, \qquad \mathcal{N}_0^{-1/2} \partial_k \Theta_{\mathrm{SR}} = \mathbf{e}_k.$$

The central anticommutator of these four constant directions is the Minkowski metric. Every second spacetime derivative of $\Theta_{\rm SR}$ vanishes, so it also solves the standard source-free constant-coefficient second-order master operator in this flat branch.

More generally, for any constant Lorentz tetrad $\bar{E}_\mu$,

$$\Theta_{\rm aff}(x) = \Theta_0 + \sqrt{\mathcal{N}_0}\,\bar{E}_\mu x^\mu$$

generates that tetrad exactly. This closes the special-relativistic representer problem, although it does not prove uniqueness or curved-space existence.

In polar coordinates, or in an accelerating frame,Γ and Ω may be nonzero even though spacetime is flat. The invariant statement of flatness is vanishing curvature, not vanishing connection coefficients in every frame.

4.5 Four covariant gradients as a tetrad

Define

$$E_\mu := \frac{1}{\sqrt{\mathcal{N}_0}} D_\mu \Theta.$$

The four objects E_0, E_1, E_2, E_3 are the local UBT rulers and clock direction. They are not four extra matter fields; they are the first covariant derivatives of the single fundamental field Θ.

Let i be the ordinary commuting complex unit and $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$ the quaternion units. In the classical Lorentz sector write

$$E_\mu = i e_\mu{}^0 \mathbf{1} + e_\mu{}^k \mathbf{e}_k, \qquad e_\mu{}^a \in \mathbb{R}.$$

Quaternion conjugation changes the signs of the three quaternion units but leaves the complex scalar coefficient untouched:

$$E_\mu^\sharp = i e_\mu{}^0 \mathbf{1} - e_\mu{}^k \mathbf{e}_k.$$

4.6 Why the anticommutator is a metric

Quaternion multiplication contains a scalar product and a cross product. In the symmetrised product the cross products cancel:

$$\boxed{\frac{1}{2}\left(E_\mu^\sharp E_\nu + E_\nu^\sharp E_\mu\right) = g_{\mu\nu}\mathbf{1}.}$$

The right-hand side contains $\mathbf{1}$, the unit biquaternion. The entire left-hand side is already central: an ordinary real number multiplied by the algebra unit. No trace, real-part map, phase projection, or fiber average is required.

Expanding the coefficients gives

$$g_{\mu\nu} = -e_\mu{}^0 e_\nu{}^0 + e_\mu{}^1 e_\nu{}^1 + e_\mu{}^2 e_\nu{}^2 + e_\mu{}^3 e_\nu{}^3,$$

which is exactly

$$g_{\mu\nu} = e_\mu{}^a e_\nu{}^b \eta_{ab}, \qquad \eta_{ab} = \mathrm{diag}(-1, 1, 1, 1).$$

The minus sign of time comes from $i^2 = -1$.

4.7 What does the other half of the product do?

The antisymmetric algebraic part is

$$\Sigma_{\mu\nu} = \frac{1}{2}\left(E_\mu^\sharp E_\nu - E_\nu^\sharp E_\mu\right).$$

It describes an oriented plane or bivector and is a natural carrier for local rotation, boost, and spin information. It is not itself the curvature.

Three expressions must be kept separate:

$$\begin{aligned} \{E_\mu, E_\nu\}_\sharp &\longrightarrow g_{\mu\nu}, \\ [E_\mu, E_\nu]_\sharp &\longrightarrow \Sigma_{\mu\nu}, \\ [D_\mu, D_\nu] &\longrightarrow \mathcal{R}_{\mu\nu}. \end{aligned}$$

4.8 Curvature from noncommuting derivatives

For a one-sided connection, the curvature has the familiar form

$$\mathcal{R}_{\mu\nu} = \partial_\mu\Omega_\nu - \partial_\nu\Omega_\mu + [\Omega_\mu, \Omega_\nu].$$

The final commutator is essential because biquaternion multiplication is noncommutative. Geometrically, curvature measures how a frame changes after transport around a small closed loop.

The equation $E_\mu = D_\mu\Theta/\sqrt{\mathcal{N}_0}$ creates an additional integrability condition. This is the point at which the representation of the connection becomes decisive.

4.9 Why the old rank problem is disappearing locally

A symmetric 4×4 metric has ten independent components. Looking only at one value of $\Theta \in \mathbb{B} \simeq \mathbb{R}^8$ suggests the misleading comparison $8 < 10$.

The metric is not built from the value of Θ alone. It is built from four covariant derivatives E_μ. In the Lorentz sector these form a real 4×4 tetrad ${e_\mu}^a$ with sixteen components. Six changes are local rotations and boosts that leave the metric unchanged. Therefore

$$16 - 6 = 10.$$

More rigorously, every small symmetric metric variation $h_{\mu\nu}$ is produced by

$$\delta {e_\mu}^a = \frac{1}{2} h_{\mu\rho} e^{\rho a}.$$

The tetrad-to-metric map therefore has rank ten at every nondegenerate tetrad. This resolves the local kinematic rank question. It does not yet prove that every curved tetrad is generated by an on-shell UBT field.

4.9.1 The right calculation can answer a poorly posed architectural question

The former single-section rank obstruction was mathematically correct in an induced-metric framework where Einstein equations were to be recovered only from normal variations of one embedding-like section. It did not constitute a no-go theorem for the covariant tetrad $E_\mu = D_\mu\Theta/\sqrt{\mathcal{N}_0}$. The compact-fiber construction repaired the former formulation by adding profile directions, but the tetrad formulation shows that this repair was not required by the original architecture.

This distinction motivates a general research rule:

> Before adding fields, modes, dimensions, projections, averages, or auxiliary structures to solve an obstruction, first test whether the obstruction is an artefact of the formulation in which it was derived.

The fiber route remains mathematically consistent in its own framework. Its problem is weak selection and representer redundancy, not an algebraic contradiction.

4.10 The problem of integrability

Once the classical connection is reconstructed from the tetrad and torsion, the defining equation becomes schematically

$$E_\mu = \mathcal{N}_0^{-1/2} \left(\partial_\mu \Theta + \rho_*(\Omega_\mu[E,T])\Theta\right).$$

The unknown E occurs on both sides. This is an implicit nonlinear first-order PDE or fixed-point system. Eliminating Ω kinematically does not by itself prove existence or uniqueness of a pair (Θ, E).

4.10.1 One-sided obstacle leading to flatness

Suppose the connection acts only from the left,

$$D^L_\mu \Theta = \partial_\mu \Theta + A_\mu \Theta,$$

so that

$$[D^L_\mu, D^L_\nu]\Theta = F^L_{\mu\nu}\Theta.$$

In a torsion-free compatible tetrad branch, antisymmetrising the covariant Hessian gives

$$F^L_{\mu\nu}\Theta = 0.$$

If Θ is invertible as a biquaternion, then

$$F^L_{\mu\nu} = 0.$$

Thus the naive one-sided regular representation cannot generically support curved GR while retaining an invertible Θ and torsion-free tetrad compatibility. Noninvertible stabilizer configurations may evade this result, but they cannot be assumed to represent arbitrary geometries.

4.10.2 Natural two-sided biquaternion derivative

Biquaternions naturally admit both left and right multiplication. Consider

$$\boxed{D_\mu \Theta = \partial_\mu \Theta + A_\mu \Theta - \Theta B_\mu.}$$

A direct expansion gives

$$\boxed{[D_\mu, D_\nu]\Theta = F^A_{\mu\nu}\Theta - \Theta F^B_{\mu\nu}.}$$

Torsion-free compatibility therefore requires

$$F^A_{\mu\nu}\Theta = \Theta F^B_{\mu\nu}.$$

For invertible Θ,

$$\boxed{F^A_{\mu\nu} = \Theta F^B_{\mu\nu}\Theta^{-1}.}$$

Unlike the one-sided case, neither curvature must vanish. The field Θ intertwines the left and right curvature representations. This is a genuine structural narrowing of GAP-10I: the two-sided/bimodule derivative is the minimal algebra-native route that avoids the generic flatness obstruction.

4.10.3 Reduction without a new field, torsion-free no-go and local right inversion with torsion

On the Lorentz slice $W_L = \{X \mid X^{\ddagger} = -X\}$, preservation of the slice and of its quadratic form reduces the pure pair, modulo a common central one-form that cancels, to

$$\boxed{A_\mu = \Omega_\mu, \qquad B_\mu = -\Omega_\mu^{\ddagger}.}$$

Thus A_μ and B_μ are not independent gravitational fields in this branch. They are the two module actions of the one spin connection.

With zero torsion, the same involution-compatible derivative makes a nondegenerate solution $D_\mu\Theta = \sqrt{\mathcal{N}_0}E_\mu$ imply

$$\boxed{\mathring{\nabla}_\mu V^\nu = \delta_\mu{}^\nu,} \qquad \mathcal{L}_V g = 2g.$$

The non-flat Schwarzschild vacuum exterior with nonzero mass has no such proper homothety. The pure pair is therefore closed kinematically and its $K = 0$ branch is closed as a torsion-free no-go.

The qualifier matters. On a sufficiently small non-null Gaussian patch, choose $\rho \neq 0$ with $g^{\mu\nu}\partial_\mu\rho\partial_\nu\rho = \varepsilon = \pm 1$ and set

$$V^\mu = \varepsilon\rho\nabla^\mu\rho, \qquad W_{\mu\nu} = g_{\mu\nu} - \mathring{\nabla}_\mu V_\nu,$$

$$\boxed{K_{\nu\mu\rho} = \frac{W_{\mu\nu}V_\rho - V_\nu W_{\mu\rho}}{V^2}.}$$

Then $K_{\nu\mu\rho} = -K_{\rho\mu\nu}$ and $\nabla_\mu^{(\mathring{\Gamma}+K)}V^\nu = \delta_\mu{}^\nu$. Therefore

$$\Theta = \sqrt{\mathcal{N}_0}V^a\mathbf{u}_a$$

obeys $D_\mu\Theta = \sqrt{\mathcal{N}_0}E_\mu$. Every smooth tetrad has such a local representer with composite metric-compatible contortion and no independent A_μ, B_μ fields.

A relative term $P_\mu X - XQ_\mu$ is no longer required for local kinematic existence. It remains a possible torsion-free completion and, if used, must be composite or auxiliary rather than a new propagating field. A common central constant does not remove the torsion-free obstruction because it cancels.

4.11 Conditional dynamic partial closures

4.11.1 Algebraic Cartan torsion equation

In the minimal first-order Hilbert–Palatini branch, independent variation of the Lorentz connection gives

$$\epsilon_{abcd}T^a \wedge e^b = \kappa\tau_{cd}.$$

For a nondegenerate tetrad this is a 24×24 pointwise linear map of rank 24. Thus zero spin current gives $T = 0$, while a specified spin current gives a unique torsion and contorsion. This closes the torsion algebra inside the minimal branch, not the derivation of that branch from fundamental UBT.

4.11.2 Propagation of the Lorentz section as a set of fixed points

The anti-linear involution

$$\mathcal{J}X = -\overline{X^{\sharp}}$$

has the physical Lorentz module as its fixed set. If a well-posed unique Cauchy evolution is $\mathcal{J}$-equivariant and its data and sources are fixed by $\mathcal{J}$, uniqueness implies that the solution remains in the Lorentz slice. The final UBT action must still be checked against these hypotheses.

4.11.3 Stability of the metric in imaginary time

A vertical flow

$$\partial_\psi e_\mu{}^a = \Lambda_\psi{}^a{}_b e_\mu{}^b, \qquad \Lambda_{\psi ab} = -\Lambda_{\psi ba},$$

is tangent to a local Lorentz gauge orbit and satisfies $\partial_\psi g_{\mu\nu} = 0$ identically. A second sufficient mechanism is a unique ψ-translation-invariant evolution with ψ-independent initial data. Selecting the physical mechanism and excluding unstable non-gauge modes remains a dynamical task.

4.11.4 Extended holonomy for prescribed curved coefficients

For prescribed E_μ, A_μ, B_μ, the inhomogeneous system

$$\partial_\mu\Theta + A_\mu\Theta - \Theta B_\mu = \sqrt{\mathcal{N}_0}E_\mu$$

becomes homogeneous parallel transport for $Y = (\Theta, 1)^T$ in an augmented bundle. A single-valued solution with initial value Y_0 exists exactly when the augmented holonomy fixes Y_0; flat augmented curvature is sufficient on a simply connected domain. This closes prescribed-coefficient integrability, not self-consistent UBT selection of the coefficients.

4.11.5 Conditional Einstein-Lambda endpoint

The minimal Palatini branch with zero spin current yields

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}.$$

Under the ordinary four-dimensional Lovelock assumptions, every local natural divergence-free second-order metric equation without extra light geometric fields has gravitational part $aG_{\mu\nu} + bg_{\mu\nu}$. Therefore the conditional low-energy endpoint is unique. UBT must still derive those assumptions, coefficients and matter terms from its canonical action.

4.12 Implicit versus transcendent equations

The words*implicit* and*transcendental* describe different properties.

An equation is implicit when the unknown occurs inside the map that is supposed to determine it:

$$E = \mathcal{F}(E, \Theta).$$

An equation is transcendental when it contains nonalgebraic functions such as an exponential, logarithm, trigonometric function, or Jacobi theta function:

$$E = e^{-E}.$$

The UBT tetrad equation is already implicit and differential. If the allowed field $\Theta(q, \tau)$ is restricted to a Jacobi-theta or another transcendental function class, then the complete system may be both implicit and transcendental. In a formal paper these properties should be stated separately, even though intuitively both make the unknown “entangled with itself.”

4.13 Derivatives versus variations

A derivative such as $D_\mu\Theta$ describes how one field configuration changes through spacetime. A variation compares nearby possible configurations:

$$\Theta_\varepsilon = \Theta + \varepsilon\,\delta\Theta.$$

The variation is

$$\delta\Theta = \left.\frac{d\Theta_\varepsilon}{d\varepsilon}\right|_0.$$

Variations are needed when deriving field equations from an action. They are not replacements for spacetime derivatives. When the connection depends on the field,

$$\delta(D_\mu\Theta) = D_\mu(\delta\Theta) + \rho_*(\delta\Omega_\mu)\Theta$$

and the induced connection variation must be included.

4.14 What is closed and what remains open

The following results are closed at their stated levels:

1. the central anticommutator metric and rank-ten tetrad map;
2. GAP-10Ω-KIN/GR: reconstruction of the compatible connection and its Levi–Civita torsion-free branch;
3. GAP-10T-PALATINI (conditional): the minimal Cartan torsion equation is algebraic and invertible;
4. GAP-10L-CONN and GAP-10L-SYM (conditional): compatible transport and unique equivariant evolution preserve the Lorentz slice;
5. GAP-10I-SR and GAP-10I-PRESCRIBED: the affine flat representer and the exact augmented-holonomy criterion for prescribed curved coefficients;
6. GAP-10I-1S: the naive one-sided invertible curved route is a conditional no-go;
7. GAP-10I-PAIR-KIN: the pure compatible pair reduces to the one spin connection;
8. GAP-10I-PAIR-GR: the $K = 0$ pure pair is a concurrent-vector no-go for the non-flat Schwarzschild vacuum exterior with nonzero mass;
9. GAP-10I-TORSION-LOCAL: every smooth tetrad has a local single-Θ representer with explicit composite metric-compatible contortion;
10. GAP-10D-PALATINI/UNIQUENESS (conditional): the minimal first-order branch and Lovelock hypotheses select Einstein–Λ;
11. GAP-10ψ-KIN/SYM: Lorentz-gauge flow or translation symmetry protects the classical metric under the stated hypotheses.

The undivided full-theory gaps are narrowed rather than fully closed:

1. GAP-10T-SPIN is closed conditionally for the direct fixed-background Palatini variation, while GAP-10T-FLAT-NOGO and GAP-10T-PAIRING-NOGO are closed as no-go results;
2. GAP-10T-DYN: compute the full composite Θ variation and derive the selected first-order branch, a canonical torsion cancellation or translational/relative completion, and normalization from canonical UBT;
3. GAP-10I-CURVED: local kinematics is closed; derive canonical action-level selection and physical admissibility of the composite torsionful branch, or an optional torsion-free relative bimodule branch, and establish regularity and global continuation of (Θ, E, A, B, T);

4. GAP-10L-DYN: prove equivariance and well-posed uniqueness for the final complete dynamics;
5. GAP-10D: derive the low-energy Palatini/Lovelock assumptions, couplings and matter action from the fundamental theory;
6. GAP-10ψ: select the stability mechanism and exclude physical unstable non-gauge modes.

GAP-B-MASTER and GAP-U2Θ remain open. The earlier compact-ψ fiber closure remains a mathematically consistent historical candidate completion, but its weak selection and representer redundancy keep it outside the minimal canonical route.

Part II

Complex time and spectral structure

Chapter 5

Continuing into complex time and Ward's identity

5.1 Why this chapter remains in the textbook

Older alpha work used a "CT scheme" built around the analytic continuation $\tau = t + i\psi$. The archived implementation is not a current source of record, but the underlying mathematics—analytic continuation, gauge identities, and the real-time limit—is standard and useful. This chapter therefore teaches those ingredients and then states exactly what they do and do not establish for UBT.

5.2 Complex time as a parameter

For an analytic quantity $F(\tau)$,

$$\tau = t + i\psi$$

allows one to move between oscillatory and damped descriptions. For an energy eigenmode, for example,

$$e^{-iE\tau/\hbar} = e^{-iEt/\hbar} e^{+E\psi/\hbar},$$

with the sign of the damping determined by the convention for the imaginary direction. In the theta convention used later,

$$e^{-\pi(\psi - it)n^2} = e^{-\pi\psi n^2} e^{i\pi t n^2}.$$

Nothing in this analytic identity by itself creates a new renormalisation constant or fixes a physical coupling.

5.3 The Ward–Takahashi identity: a standard argument

In QED let $S(p)$ be the full fermion propagator and $\Gamma^\mu(p+q, p)$ the full electromagnetic vertex. Gauge invariance gives the Ward–Takahashi identity

$$q_\mu \Gamma^\mu(p+q, p) = S^{-1}(p+q) - S^{-1}(p).$$

Taking the limit $q \to 0$ yields

$$\Gamma^\mu(p, p) = \frac{\partial S^{-1}(p)}{\partial p_\mu}.$$

At the level of renormalisation constants this enforces

$$\boxed{Z_1 = Z_2,}$$

provided the regulator and subtraction prescription preserve gauge symmetry. This is a standard gauge-theory result. A complex-time continuation may use it only after demonstrating that its contour/regulator preserves the hypotheses required by the identity.

5.4 The transversality of the photon's own energy

The same gauge symmetry implies

$$q_\mu \Pi^{\mu\nu}(q) = 0,$$

so the vacuum polarisation has the form

$$\Pi^{\mu\nu}(q) = \left(q^2 g^{\mu\nu} - q^\mu q^\nu\right) \Pi(q^2).$$

(up to convention). This is why charge renormalisation can be expressed in terms of the transverse photon two-point function. Again, this is a standard field-theory constraint, not a UBT-specific theorem.

5.5 Controlled UBT bridge

A UBT complex-time renormalisation construction would need to establish, for its actual action and contour:

1. analyticity sufficient to continue between the chosen ψ contour and the real-time slice;
2. preservation of the relevant gauge/BRST identities;
3. a regulator and counterterm scheme compatible with those identities;
4. continuity to standard QED in the appropriate real-time limit;
5. an explicit calculation of any finite CT-specific remainder rather than setting it by definition.

Only after these steps could a CT-specific factor be promoted from a modelling choice to a derived quantity.

5.6 Why the archive text $\mathcal{R}_{\mathrm{UBT}} = 1$ is not downloaded

The historical alpha branch stated a two-loop baseline $\mathcal{R}_{\mathrm{UBT}} = 1$ under assumptions labelled A1–A3 and then called the resulting alpha baseline "fit-free". Current Tier-A status documents no longer allow that wording to stand as a completed alpha derivation: the coefficient bridge G137-B is open and the effective multiplicity has an audit gap. Therefore the live textbook keeps the Ward-identity mathematics but does not reuse the archived theorem as a current source of record.

This is a useful example of a general rule: a correct conditional field-theory identity can survive even when a larger phenomenological interpretation built on top of it is downgraded.

5.7 Link to Theta Chapter

The reduced theta kernel

$$S(t, \psi) = \sum_n a_n e^{-\pi\psi n^2} e^{i\pi t n^2}$$

provides a cleaner controlled laboratory for complex time than the old alpha renormalisation route. The same parameter ψ acts as a heat-time variable, while t carries the oscillatory phase. Fourier, Laplace, Mellin, and Poisson/Jacobi transforms can then be derived exactly before any physical UBT claim is attached to them.

Chapter 6

Theta, Mellin, zeta, paths and prime structure

6.1 Why this chapter exists

This chapter is deliberately broader than a canonical UBT claim document. Its purpose is pedagogical: to place several classical mathematical structures next to the reduced UBT theta bridge so that their relations can be seen in one place. A result does*not* become a UBT theorem merely because it is useful inside UBT.

We therefore use three status labels in the prose:

- **Classical**: standard mathematics or standard quantum mechanics;
- **Bridge**: an exact statement about the reduced theta model used to connect standard mathematics to UBT notation;
- **UBT open**: a statement that would have to be derived from the canonical UBT field equations or action before it could become a physical UBT claim.

The central reduced object is

$$S(t, \psi) = \sum_{n \geq 1} a_n \, e^{-\pi \psi n^2} e^{i\pi t n^2} = \sum_{n \geq 1} a_n e^{-\pi(\psi - it)n^2}. \tag{6.1}$$

It is the same reduced complex-time kernel used in `canonical/bridges/theta_complex_time_bridge.tex`. In this chapter it is a **bridge model**; the canonical UBT master field $\Theta(q, \tau)$ is not assumed to be a Jacobi theta function.

Sign convention used in related RH notes. Some earlier UBT/RH scratch work used a positive heat variable $y = -\psi$ and therefore wrote the theta parameter as $t - i\psi = t + iy$ with convergence on the corresponding $y > 0$ half-line. The present chapter writes the damping variable directly as $\psi > 0$ in Eq. (6.1). The transform identities are identical after the relabelling $y = -\psi$; the sign convention must be declared before a physical UBT interpretation is attached to either half-line.

6.2 One spectrum, several representations

6.2.1 Orthogonality of modes versus nonorthogonality of theta templates

For fixed ψ, the elementary time modes in Eq. (6.1) are

$$e^{i\pi n^2 t}. \tag{6.2}$$

On the interval $t \in [0,2]$ they are orthogonal:

$$\int_0^2 e^{i\pi(n^2-m^2)t}\,dt = 2\,\delta_{nm}, \qquad n,m \geq 1. \tag{6.3}$$

The important distinction is that the*complete templates* $S_\psi(t) := S(t,\psi)$ at different values of ψ are generally not orthogonal.

Theorem 6.1 (Classical Gram kernel identity)**.** *For the inner product*

$$\langle f,g\rangle := \frac{1}{2}\int_0^2 f(t)\overline{g(t)}\,dt, \tag{6.4}$$

and amplitudes S_ψ of Eq. (6.1),

$$G(\psi,\phi) := \langle S_\psi, S_\phi\rangle = \sum_{n\geq 1} |a_n|^2 e^{-\pi(\psi+\phi)n^2}. \tag{6.5}$$

For $a_n \equiv 1$,

$$G(\psi,\phi) = \frac{\vartheta_3(0 \mid i(\psi+\phi)) - 1}{2}. \tag{6.6}$$

Proof. Insert the two sums in the inner product. Equation (6.3) eliminates all $m \neq n$ cross terms, leaving

$$\sum_n |a_n|^2 e^{-\pi\psi n^2} e^{-\pi\phi n^2}.$$

For unit weights, the standard identity $\vartheta_3(0 \mid iu) = 1 + 2\sum_{n\geq 1} e^{-\pi u n^2}$ gives Eq. (6.6). □

Practical consequence. Template non-orthogonality is not an obstacle to detection. If an observed signal is $f(t)$, a normalized matched-filter score is

$$\rho(\psi,\Delta) = \frac{\left|\int f(t)\overline{S(t-\Delta,\psi)}\,dt\right|^2}{\|f\|^2\,\|S_\psi\|^2}. \tag{6.7}$$

The best single template is obtained by maximizing ρ over (ψ,Δ). Orthogonality becomes important when many templates are fitted simultaneously; then the Gram matrix must be inverted or regularized, for example through $G + \lambda I$.

6.2.2 Real-time Fourier transform

Take the Fourier convention

$$\widehat{f}(\omega) = \int_{-\infty}^{\infty} f(t)e^{-i\omega t}\,dt. \tag{6.8}$$

Formally, or distributionally,

$$\widehat{S}(\omega,\psi) = 2\pi \sum_{n\geq 1} a_n e^{-\pi\psi n^2}\delta(\omega - \pi n^2). \tag{6.9}$$

Thus the quadratic mode spectrum is already a Dirac comb in real-time frequency; no mode-by-mode demodulation is needed to create the delta lines.

6.2.3 Laplace transform in imaginary time

For $\operatorname{Re} z$ sufficiently large,

$$\mathcal{L}_\psi[S](z;t) = \int_0^\infty e^{-z\psi} S(t,\psi)\,d\psi \tag{6.10}$$

$$= \sum_{n\geq 1} \frac{a_n e^{i\pi t n^2}}{z + \pi n^2}. \tag{6.11}$$

The same eigenvalues that gave Fourier delta lines now appear as resolvent poles $z = -\pi n^2$.

6.2.4 Mellin's transformation in imaginary time

For $\operatorname{Re} s$ in the absolute-convergence half-plane,

$$\mathcal{M}_\psi[S](s;t) := \int_0^\infty \psi^{s-1} S(t,\psi)\, d\psi \tag{6.12}$$

$$= \Gamma(s)\pi^{-s} \sum_{n\geq 1} \frac{a_n e^{i\pi t n^2}}{n^{2s}}. \tag{6.13}$$

The identity follows from

$$\int_0^\infty \psi^{s-1} e^{-\pi n^2 \psi}\, d\psi = \Gamma(s)(\pi n^2)^{-s}. \tag{6.14}$$

For the Gram kernel with $a_n \equiv 1$ and $u = \psi + \phi$,

$$\int_0^\infty u^{s-1} G(u)\, du = \Gamma(s)\pi^{-s}\zeta(2s). \tag{6.15}$$

This is classical theta–Mellin mathematics. Its relevance here is organizational: the same reduced theta bank that causes non-orthogonal templates is also a heat trace whose Mellin representation is a spectral zeta function.

The spectral triangle. Equations (6.9),(6.11), and (6.13) give three views of the same quadratic spectrum:

$$\boxed{\begin{array}{lcl} \mathcal{F}_t & \longrightarrow & \delta(\omega - \pi n^2), \\ \mathcal{L}_\psi & \longrightarrow & (z + \pi n^2)^{-1}, \\ \mathcal{M}_\psi & \longrightarrow & \Gamma(s)\pi^{-s} n^{-2s}. \end{array}} \tag{6.16}$$

6.3 Mellin is the Fourier transform in logarithmic coordinate

Let

$$M_f(s) = \int_0^\infty f(x) x^{s-1}\, dx, \qquad s = \sigma + i\omega. \tag{6.17}$$

Set $x = e^u$, so $dx = e^u du$. Then

$$M_f(\sigma + i\omega) = \int_{-\infty}^\infty f(e^u) e^{\sigma u} e^{i\omega u}\, du. \tag{6.18}$$

Apart from the conventional sign of the Fourier phase, this is a Fourier transform in $u = \ln x$ of the weighted signal $f(e^u)e^{\sigma u}$. Thus Mellin analysis is the natural harmonic analysis for multiplicative scaling, whereas ordinary Fourier analysis is natural for additive translation.

This observation is especially useful when interpreting zeta functions: zeta's vertical variable $\operatorname{Im} s$ is a log-scale frequency.

6.4 Current cut and time-phased family of zeta functions

UBT coordinate convention. In the working complex-time interpretation used in this research thread, the present slice is chosen as $t = 0$. This is a UBT coordinate convention, not a statement that the Riemann zeta function intrinsically knows a physical "present."

For unit weights $a_n \equiv 1$, define the quadratic time twist

$$Z_\Theta(s;t) := \sum_{n=1}^\infty \frac{e^{i\pi t n^2}}{n^{2s}}. \tag{6.19}$$

Then Eq. (6.13) becomes

$$\boxed{\mathcal{M}_\psi[S](s;t) = \Gamma(s)\pi^{-s} Z_\Theta(s;t).} \tag{6.20}$$

At the present slice,

$$\boxed{Z_\Theta(s;0) = \zeta(2s).} \tag{6.21}$$

Thus ordinary zeta is one member—the $t = 0$ member—of a natural family generated by the same reduced complex-time theta kernel.

Remark 6.2. The exponential factor in Eq. (6.19) is an*additive quadratic twist*. For generic t the resulting Dirichlet series does not have the ordinary Euler product. One must not infer prime-factor dynamics merely from the existence of the $t = 0$ Euler product.

6.4.1 Simple special times

The family is periodic under $t \mapsto t + 2$ because $n^2 \in \mathbb{Z}$:

$$Z_\Theta(s; t + 2) = Z_\Theta(s; t). \tag{6.22}$$

At $t = 1$,

$$Z_\Theta(s;1) = \sum_{n\geq 1} \frac{(-1)^{n^2}}{n^{2s}} = \sum_{n\geq 1} \frac{(-1)^n}{n^{2s}} \tag{6.23}$$

$$= -\eta(2s) = -(1 - 2^{1-2s})\zeta(2s), \tag{6.24}$$

where η is the Dirichlet eta function.

6.4.2 Rational times: decomposition using the Hurwitz zeta function

Let $t = a/q$ with integers a and $q > 0$. The coefficient

$$c_n = e^{i\pi a n^2/q} \tag{6.25}$$

is periodic with period $M = 2q$(sometimes the true period is smaller). Splitting the Dirichlet series into residue classes gives the exact classical identity

$$\boxed{Z_\Theta\left(s; \frac{a}{q}\right) = M^{-2s} \sum_{r=1}^{M} e^{i\pi a r^2/q} \zeta\left(2s, \frac{r}{M}\right), \qquad M = 2q,} \tag{6.26}$$

where $\zeta(s, \alpha)$ is the Hurwitz zeta function.

Proof. Write every $n \geq 1$ uniquely as $n = mM + r$ with $m \geq 0$ and $1 \leq r \leq M$. Periodicity gives $c_{mM+r} = c_r$, and

$$Z_\Theta(s; a/q) = \sum_{r=1}^{M} c_r \sum_{m=0}^{\infty} (mM + r)^{-2s}$$

$$= M^{-2s} \sum_{r=1}^{M} c_r \zeta\left(2s, \frac{r}{M}\right).$$

□

This makes the "alternative zeta functions" generated away from $t = 0$ precise. The ordinary zeta, eta, Hurwitz zeta combinations, and quadratic Gauss sums are not unrelated objects here: they are different slices or representations of one quadratic-phase family.

6.5 Rational Revivals and Quadratic Gaussian Sums

At rational $t = a/q$, the phase $e^{i\pi a n^2/q}$ depends only on a finite residue class. The corresponding finite quadratic sums are Gauss sums. A common normalization is

$$g(a, q) = \sum_{r=0}^{q-1} e^{2\pi i a r^2/q}. \tag{6.27}$$

For coprime moduli q_1, q_2, the Chinese remainder theorem gives, with the normalization in Eq. (6.27),

$$g(a, q_1 q_2) = g(a q_2, q_1)\, g(a q_1, q_2), \qquad \gcd(q_1, q_2) = 1. \tag{6.28}$$

Thus composite denominators factor into coprime pieces, and prime powers are the irreducible arithmetic building blocks of rational-time revival structure.

This is a more promising place to search for a*dynamical* origin of UBT prime sectors than simply observing the Euler product of $\zeta(2s)$, because the Gauss-sum arithmetic is carried directly by the quadratic theta phase.

6.6 Sum of theta modes and Feynman sum of windings

The theta/path relation is exact in a controlled textbook example: a free particle of mass m moving on a circle of circumference L.

The momentum eigenfunctions and energies are

$$\phi_n(x) = L^{-1/2} e^{2\pi i n x/L}, \qquad E_n = \frac{\hbar^2}{2m}\left(\frac{2\pi n}{L}\right)^2. \tag{6.29}$$

Hence the real-time propagator is

$$K(x, x'; T) = \frac{1}{L} \sum_{n\in\mathbb{Z}} \exp\!\left(\frac{2\pi i n(x - x')}{L} - \frac{i E_n T}{\hbar}\right). \tag{6.30}$$

This is a Jacobi-theta mode sum. Poisson summation transforms it into

$$\boxed{K(x, x'; T) = \sqrt{\frac{m}{2\pi i \hbar T}} \sum_{w\in\mathbb{Z}} \exp\left[\frac{i m (x - x' + wL)^2}{2\hbar T}\right].} \tag{6.31}$$

The integer w labels winding sectors. The exponent is the classical free-particle action for the lifted path from x' to $x + wL$:

$$S_w^{\mathrm{cl}} = \frac{m(x - x' + wL)^2}{2T}. \tag{6.32}$$

Therefore the same propagator admits two exact representations:

$$\boxed{K_\Sigma \quad \Longleftrightarrow \quad K_\mathbb{Z}} \tag{6.33}$$

Here K_Σ denotes the spectral mode sum and $K_\mathbb{Z}$ the sum over winding sectors, with Poisson/Jacobi transformation as the bridge.

Important qualification. The classical trajectory is selected by*stationary* action, $\delta S = 0$, not necessarily minimum action. Several stationary paths are ordinary quantum mechanics; by themselves they do not imply a multiverse interpretation.

6.7 Where prime numbers come in and where they don't

6.7.1 Euler's product on an unphased section

At $t = 0$ and unit weights,

$$Z_\Theta(s;0) = \zeta(2s) = \prod_p \left(1 - p^{-2s}\right)^{-1}, \qquad \operatorname{Re} s > \frac{1}{2}. \tag{6.34}$$

This is a profound change of representation from an additive sum over all positive integers to a multiplicative product over primes. It is nevertheless a classical property of the integers, not by itself evidence that UBT dynamics selects prime modes. For general weights a_n, the Dirichlet series $\sum |a_n|^2 n^{-2s}$ has an Euler product only when the coefficients satisfy the appropriate multiplicativity conditions.

6.7.2 Grids of poles of local Euler factors

A single formal local factor

$$L_p(s) = \left(1 - p^{-2s}\right)^{-1} \tag{6.35}$$

has poles when $p^{-2s} = 1$. Writing $s = \sigma + i\omega$ gives

$$\sigma = 0, \qquad \omega_{p,k} = \frac{\pi k}{\ln p}, \qquad k \in \mathbb{Z}. \tag{6.36}$$

For distinct primes $p \neq q$, nonzero pole frequencies cannot coincide. If $k/\ln p = \ell/\ln q$ with nonzero integers k, ℓ, then $q^k = p^\ell$, contradicting unique factorization. All local factors share the trivial $k = 0$ location.

These are poles of the*individual formal Euler factors*. They lie outside the Euler-product convergence half-plane and must not be confused with poles of the analytically continued global $\zeta(2s)$, whose pole is at $s = 1/2$.

If $N(T;P)$ counts positive local-factor pole frequencies up to height T for primes $p \leq P$, then

$$N(T;P) = \sum_{p \leq P} \left\lfloor \frac{T \ln p}{\pi} \right\rfloor \tag{6.37}$$

$$= \frac{T}{\pi}\vartheta(P) + O(\pi(P)), \tag{6.38}$$

where

$$\vartheta(P) = \sum_{p \leq P} \ln p \tag{6.39}$$

is Chebyshev's theta function. The prime number theorem implies $\vartheta(P) \sim P$.

6.7.3 Vertical Fourier spectrum and powers of prime numbers

At fixed $\sigma > 1/2$,

$$\zeta\big(2(\sigma + i\omega)\big) = \sum_{n \geq 1} n^{-2\sigma} e^{-i(2\ln n)\omega}. \tag{6.40}$$

Thus Fourier transformation in ω produces spectral lines at

$$\xi_n = 2\ln n. \tag{6.41}$$

To isolate primes and prime powers, use the logarithmic derivative:

$$-\frac{\zeta'(z)}{\zeta(z)} = \sum_{n \geq 1} \frac{\Lambda(n)}{n^z} = \sum_p \sum_{k \geq 1} (\ln p) p^{-kz}, \qquad \operatorname{Re} z > 1. \tag{6.42}$$

With $z = 2(\sigma + i\omega)$,

$$-\frac{\zeta'(2(\sigma+i\omega))}{\zeta(2(\sigma+i\omega))} = \sum_p \sum_{k\geq 1} (\ln p) p^{-2k\sigma} e^{-i(2k\ln p)\omega}. \tag{6.43}$$

The prime-power frequency comb therefore sits at

$$\boxed{\xi_{p,k} = 2k\ln p.} \tag{6.44}$$

Distinct prime-power lines coincide only when the prime powers themselves are identical.

6.7.4 Counting the primes and zero frequencies of the zeta function

The weighted prime-power counting function

$$\psi_C(x) = \sum_{n\leq x} \Lambda(n) \tag{6.45}$$

is related by the classical explicit formula to the nontrivial zeros $\rho = \beta + i\gamma$ of ζ. Schematically,

$$\psi_C(x) = x - \sum_\rho \frac{x^\rho}{\rho} + \cdots . \tag{6.46}$$

Writing $x = e^u$ turns each zero contribution into

$$x^\rho = e^{\beta u} e^{i\gamma u}. \tag{6.47}$$

Thus the imaginary parts γ of zeta zeros act as oscillation frequencies in the logarithmic variable $u = \ln x$. This is the classical primes–zeros spectral bridge.

6.8 What this means for UBT prime sectors

Two distinct mechanisms must not be conflated.

Route A: trace/orbit route. Trace-formula analogies suggest that primes should correspond to primitive periodic objects whose repeated traversals generate prime powers. For the ordinary free particle on S^1, however, the primitive-orbit length spectrum is not $\{\ln p\}$. Therefore

> **GAP-THETA-PRIME-1:** derive from a genuine UBT operator a primitive-orbit or spectral structure whose natural primitive data are $\ln p$, or do not claim that the Euler product dynamically derives UBT prime sectors.

Route B: rational-revival route. The quadratic theta dynamics itself has arithmetic structure at rational times. Gauss sums factor through coprime denominators, making prime powers natural building blocks. This motivates

> **GAP-THETA-PRIME-2:** rewrite the existing UBT α prime-sector construction in revival/winding variables and test whether its distinguished primes arise from rational-time Gauss-sum structure rather than from a post-selected list.

These routes are compatible as research directions but are not yet known to be the same mechanism.

6.9 Operator question behind the whole picture

Equation (6.1) can be written formally by introducing

$$H_0|n\rangle = \pi n^2|n\rangle. \tag{6.48}$$

Then its mode evolution is generated by

$$U_0(t,\psi) = e^{-\psi H_0} e^{+itH_0}. \tag{6.49}$$

This makes the reduced theta amplitude genuinely propagator-like. The decisive UBT question is stronger:

> **GAP-THETA-PROP:** derive an operator H_Θ from the canonical UBT action or field equations such that an appropriate kernel or matrix element of $e^{-\psi H_\Theta} e^{\pm itH_\Theta}$ yields the theta structure used here.

Until this gap is closed, the mode/winding, Mellin/zeta, and revival/prime relations are rigorous mathematics of the reduced bridge model, not a derivation of UBT dynamics.

6.10 Big picture map

The useful synthesis is therefore

$$\boxed{\begin{array}{ccc}
 & \mathcal{K}_\Theta & \\
 & \displaystyle\sum_n a_n e^{-\pi\psi n^2} e^{i\pi t n^2} & \\
\swarrow & \downarrow & \searrow \\
\mathcal{F}_t/\mathcal{L}_\psi & \mathcal{M}_\psi & \mathcal{P} \\
\mathcal{D} & Z_\Theta(s;t) & \mathcal{W} \\
 & \downarrow\ t=0 & \downarrow\ t\in\mathbb{Q} \\
 & \zeta(2s) & \mathcal{G}_\mathbb{Q} \\
 & \downarrow & \\
 & \mathcal{E}_p &
\end{array}} \tag{6.50}$$

In the diagram, $\mathcal{K}_\Theta$ is the reduced theta kernel, $\mathcal{P}$ is the Poisson/Jacobi transform, $\mathcal{D}$ denotes the delta-line/resolvent branch, $\mathcal{W}$ the winding/path sum, $\mathcal{G}_\mathbb{Q}$ rational-time Gauss arithmetic, and $\mathcal{E}_p$ the Euler-product, prime-power, and zero branch. The classical arrows are exact. The research task is to determine which of them survive when the reduced kernel is replaced by a canonical UBT operator and whether either prime route explains the existing UBT α sectors.

Part III

Phenomenology, verification and applications

Chapter 7

The fine structure constant program: what is derived and what is not

7.1 Status before algebra

The authoritative repository status is deliberately stronger than any elegant formula in this chapter:

> The physical fine-structure constant is not yet derived from first principles in UBT. The integer-137 route is structural/conditional, while the bridge fixing the required effective coefficient from the UBT action remains open.

The live sources of status are `STATUS.md` and `WHAT_IS_PROVED.md`. The alpha ledger is `canonical/alpha/`
`ALPHA_MASTER_STATUS.md`. Historical texts using the phrase "fit-free alpha derivation" do not override these ledgers.

7.2 Conditional winding potential

A central reduced model studies

$$\boxed{V_{\text{eff}}(n) = n^2 - Bn\ln n, \qquad n > 0.}$$

Treating n temporarily as a continuous variable,

$$\frac{dV_{\text{eff}}}{dn} = 2n - B(\ln n + 1).$$

A stationary point therefore satisfies

$$\boxed{B = \frac{2n}{\ln n + 1}.}$$

If one asks for a stationary point at $n = 137$, the required coefficient is

$$B_{137} = \frac{274}{\ln 137 + 1} \approx 46.284.$$

This equation is an exact property of the reduced potential. It does *not* explain why the UBT action must generate that value of B.

The second derivative is

$$\frac{d^2V_{\text{eff}}}{dn^2} = 2 - \frac{B}{n}.$$

At a stationary point this becomes

$$2 - \frac{2}{\ln n + 1} = \frac{2\ln n}{\ln n + 1} > 0 \qquad (n > 1),$$

so the continuous stationary point is a local minimum. Integer and prime-sector restrictions can then be studied as an additional discrete selection problem.

7.3 What "prime stability" can and cannot mean

The repository contains a structural prime-stability analysis in which the integer sector around 137 is examined under a specified selection rule. A prime-labelled minimum can be a reproducible result of that reduced model. However:

- the primality of 137 is standard number theory;
- selection of a prime sector does not by itself identify α^{-1} with that sector;
- the physical correction from 137 to 137.036 . . . needs a separate derivation;
- the coefficient B must still be obtained from the UBT dynamics rather than inferred from the target value.

The theta/Mellin/Feynman chapter develops possible mechanisms for understanding where multiplicative prime structure could enter, but those mechanisms are research bridges, not completed alpha proofs.

7.4 Audit N_{eff}: two different counts must not be confused

A previous derivation used

$$N_{\text{eff}} = 3 \times 2 \times 2 = 12.$$

The current audit distinguishes at least two objects:

$$\begin{aligned} N_{\text{eff}}^{\text{twist}} &= 12, && \text{(T)}, \\ N_{\text{eff}}^{\text{loop}} &= 3, && \text{(L)}. \end{aligned}$$

Here T denotes the SU(2)-twist construction and L the direct audited scalar-action loop count. The identification

$$N_{\text{eff}}^{\text{loop}} = N_{\text{eff}}^{\text{twist}}$$

is not presently a proved theorem. Therefore a formula that inserts 12 into a loop coefficient must name the twist assumptions; it cannot be described as a unique consequence of $\mathbb{C} \otimes \mathbb{H}$ alone.

Correspondingly, the repository records two baseline coefficients in different routes,

$$B_0^{\text{loop}} = 2\pi, \qquad B_0^{\text{twist}} = 8\pi.$$

with the bridge from either baseline to the phenomenologically required $B \approx 46.284$ still open.

7.5 Gap G137-B

The decisive alpha problem can be stated without rhetoric:

> Starting from the canonical UBT action and its justified field content, derive the effective reduced coefficient multiplying $n \ln n$, including all normalisations and multiplicities, and show that the derivation is independent of the measured value of α.

This is Gap G137-B. Several historical and research-track routes have produced interesting modular, eta, Gamma, determinant, and winding structures, but the current master status records the required insertion into the physical coefficient as open.

7.6 Why it still makes sense to study the theta/Mellin branch

The failure of the naive chain

$$\Theta \to \mathcal{M} \to \zeta \to \mathcal{E} \to \mathcal{P}$$

where $\mathcal{M}$, $\mathcal{E}$, and $\mathcal{P}$ denote the Mellin transform, Euler product, and prime sectors, respectively, to*dynamically* select primes is itself useful. The Euler product of a standard theta Mellin transform reflects multiplicative arithmetic already present in the integer index set. A stronger mechanism would have to identify primitive sectors dynamically.

Rational-time theta revivals are one candidate because quadratic Gauss sums factor over coprime denominators. In that setting prime powers are genuinely irreducible building blocks of the revival arithmetic. Whether the alpha prime-sector construction can be rewritten in those variables is an explicit open question, not an established result.

7.7 Safe summary

The statements to remember are:

1. the reduced stationary equation and its stability calculus are exact;
2. the integer/prime structural selection can be studied reproducibly;
3. B is not yet derived from the UBT action at the required value;
4. the relevant multiplicity bookkeeping is not fully reconciled;
5. $\alpha^{-1} = 137.036\ldots$ is not a first-principles UBT result today.

This is intentionally less dramatic than older archive text and more useful for future work, because it tells us exactly which equation must be closed next.

Chapter 8

Tests, reproducibility and formal verification

The live verification layer is the active repository, not `ARCHIVE/`. Important entry points are:

- `tests/` for scientific, provenance and architectural regression tests;
- `tools/apply_provenance_headers.py` for AI provenance marking;
- `tools/latex_audit.py` for isolated translations of LaTeX documents;
- `tools/check_alpha_claims.py` and active alpha tests for review of claims sensitive to current status;
- numerical checks of theta/Mellin/Gaussian branches are provided by `research_tracks/theta_spectral/theta_zeta_probe.py`;
- `docs/DERIVATION_VERIFICATION_POLICY.md` as binding rules derivation checks in papers.

8.1 Four different types of control

It is important not to confuse the different levels of verification. LaTeX translation only proves that the document is syntactically composable. The numerical test verifies the specific numerical consequences in a given accuracy. A CAS tool, such as SymPy or Maxima, it verifies the token identity we actually gave it. Formal system Lean checks whether a formalized conclusion follows from formalized assumptions. Neither of these checks alone proves that the relevant assumptions are physically selected by canonical UBT dynamics.

8.2 Lean as the preferred formal controller

It is the preferred tool for new or substantially changed theorem derivations Lean. If the statement is reasonably formalizable, the goal is to have alongside the analytic derivation also compiling `.lean` proof. If there is no formalization yet done or is not available in a given Lean environment, the status must be explicit marked as `LEAN-PENDING`; one must not write that the statement is "formally proven" just because the agent generated the Lean syntax.

Lean does not replace other controls. For the main result of the paper, the second is desirable, independent pipeline: for example SymPy/Maxima for symbolic algebra and NumPy/Octave/MATLAB/Mathcad for numerical reproduction. Two implementations should if possible use different wording so as not to repeat the same mistake.

8.3 Mandatory record of paper verification

Each new or substantially changed paper should contain a section *Verification* or a link to an accompanying record. For the main statement, at least:

- statement, equation, or lemma identifier;
- analytical status and assumptions;
- used tool and its version;
- path to reproduction script, test, worksheet or `.lean` file;
- result and possible tolerance;
- exactly what the check verifies and what it does not;
- Lean formalization status: `PROVED`, `PARTIAL`, `LEAN-PENDING`, or `NOT-APPLICABLE`.

8.4 CI textbook

The dedicated workflow is `.github/workflows/textbook.yml`. It runs on relevant pushes and pull requests and can also be started manually. It first checks bilingual parity and then uses `tools/latex_audit.py -strict`. A failure in any standalone textbook document therefore fails the workflow.

After the strict audit, the workflow calls the same Makefile used locally and creates exactly two official PDFs in `build/textbook/public/`:

- `UBT_Textbook_EN.pdf` — English edition;
- `UBT_Textbook_CS.pdf` — Czech edition.

Each PDF is uploaded as a separately named GitHub Actions artifact. Email delivery attaches both editions and is allowed only after a successful push to `master`; pull requests and manual runs never send email.

For local published copy can be run

```
make -C docs/textbook publish
```

which copies the same pair into `docs/pdfs/`. Generated PDFs are only publication outputs; scholarly status remains governed by source text and the provenance/status registry.

8.5 Minimal local check before textbook patch

The recommended sequence is:

```
python tools/apply_provenance_headers.py --apply
python tools/apply_provenance_headers.py --check
python tools/latex_audit.py --include 'docs/textbook/*.tex' \
  --include 'docs/textbook/**/*.tex' --jobs 2 --timeout 240 --strict
make -C docs/textbook verify
```

A successful build confirms syntactic and reproduction properties. It does not increase itself from self-Tier-C claims on proven UBT result.

Chapter 9

Applications and short-term predictions

Part IV

Reference appendices

Appendix A

Formal proofs and lemmas

This appendix collects three short arguments used repeatedly in the textbook. They are included here so that the main chapters can remain readable without turning standard algebra into an external assumption.

A.1 Central anticommutator on the Lorentz slice

Let

$$E_\mu = i e_\mu{}^0 \mathbf{1} + e_\mu{}^k \mathbf{e}_k, \qquad E_\mu^\sharp = i e_\mu{}^0 \mathbf{1} - e_\mu{}^k \mathbf{e}_k.$$

Using

$$\mathbf{e}_j \mathbf{e}_k = -\delta_{jk} \mathbf{1} + \epsilon_{jk\ell} \mathbf{e}_\ell$$

the antisymmetric cross-product terms cancel after symmetrisation. Hence

$$\frac{1}{2}\left(E_\mu^\sharp E_\nu + E_\nu^\sharp E_\mu\right) = \left(-e_\mu{}^0 e_\nu{}^0 + \sum_{k=1}^{3} e_\mu{}^k e_\nu{}^k\right) \mathbf{1}.$$

The coefficient is the tetrad metric $g_{\mu\nu} = e_\mu{}^a e_\nu{}^b \eta_{ab}$. No trace or real-part projection is required.

A.2 Rank of the tetrad-to-metric map

For

$$g_{\mu\nu} = e_\mu{}^a e_\nu{}^b \eta_{ab}$$

the differential is

$$\delta g_{\mu\nu} = \eta_{ab}\left(\delta e_\mu{}^a e_\nu{}^b + e_\mu{}^a \delta e_\nu{}^b\right).$$

At a nondegenerate tetrad, every symmetric tensor $h_{\mu\nu}$ is obtained by

$$\delta e_\mu{}^a = \frac{1}{2} h_{\mu\rho} e^{\rho a}.$$

The image therefore has dimension ten. The kernel consists of the six infinitesimal Lorentz transformations $\delta e_\mu{}^a = \lambda^a{}_b e_\mu{}^b$ with $\lambda_{ab} = -\lambda_{ba}$.

A.3 Why variations are not derivatives

A spacetime derivative and a field-space variation act in different directions:

$$D_\mu \Theta = \partial_\mu \Theta + \rho_*(\Omega_\mu)\Theta, \qquad \Theta_\varepsilon = \Theta + \varepsilon\, \delta\Theta.$$

When the connection depends on the field, the chain rule gives

$$\delta(D_\mu\Theta) = D_\mu(\delta\Theta) + \rho_*(\delta\Omega_\mu)\Theta.$$

Omitting the second term changes the composite Euler–Lagrange system. This is why a fixed-connection calculation cannot automatically be reused after a composite connection has been substituted.

Appendix B

FAQ for reviewers and engineers

B.1 Is the metric obtained by projecting a biquaternion?

No. On the classical Lorentz slice the symmetrised product

$$\frac{1}{2}(E^\sharp_\mu E_\nu + E^\sharp_\nu E_\mu) = g_{\mu\nu}\mathbf{1}$$

is already central. A trace, real-part prescription, preferred phase section or fibre average is not part of the canonical local definition.

B.2 Does one eight-real-component field have too few components for a metric?

The metric is constructed from four covariant first derivatives, not from the value of Θ alone. Their Lorentz-real components form a tetrad with sixteen entries. After the six local Lorentz directions are removed, the tetrad-to-metric differential has rank ten.

B.3 Is the connection an additional fundamental field?

Not in the locked classical architecture. A nondegenerate tetrad and specified torsion determine the metric-compatible connection. What remains open is the dynamical law selecting torsion and the precise composite or auxiliary realisation required by the final action.

B.4 Has Einstein dynamics been derived unconditionally?

No. Local kinematic representability and several conditional Palatini results are established, but microscopic selection and normalisation of the complete action remain open. The status must not be promoted from conditional closure without that derivation.

B.5 How can a calculation be reproduced?

Start from the cited active source, record its assumptions and claim tier, run the named verifier and its pytest wrapper, and distinguish algebraic checks from physical selection. A successful CAS or Lean check verifies only the statement that was encoded.

Appendix C

Speculative extensions (read separately)

C.1 Boundary of this appendix

This appendix is a map of research directions, not a source of canonical axioms. None of the items below may be used to close a gap in the main theory unless its assumptions, equations and reduction map are derived independently.

C.2 Complex-time and physical phase questions

The canonical coordinate is $\tau = t + i\psi$. Possible compactness, thermal periodicity or dynamical stabilisation of the ψ direction remains a research question. A preferred section or fibre average must not be inserted into the local metric definition merely to simplify observation.

C.3 Topological and solitonic sectors

Hopfion-like, winding and defect solutions can be studied as candidate sectors of a specified action. Their topology alone does not determine particle masses, couplings or stability. Each proposal needs an existence result, a finite-energy condition and a fluctuation analysis.

C.4 Consciousness-related models

Models that associate phase-space or stochastic structures with consciousness belong to the speculative research layer. They are not consequences of the canonical UBT axioms and currently carry no established empirical bridge.

C.5 Promotion checklist

A speculative construction can enter an active physical track only after it states a falsifiable observable, derives its variables from the single-field architecture, preserves the locked geometry, and passes the repository's verification and bilingual-review gates.